

Bundle 5 — Runpod training / inference / campaign execution loop

Scrum-master view

Bundle objective: prepare the remote execution layer for expensive NCA-ART-GRN research runs.

Bundle 3 prepared the **single-candidate mechanism loop**.

Bundle 4 prepared **search and comparison tools**.

Bundle 5 prepares the same work to run safely on Runpod or other remote compute.

This bundle does **not** invent new science. It scales the science from Bundles 3 and 4.

Bundle 3

evaluates one candidate mechanism deeply

Bundle 4

compares many candidates and search methods

Bundle 5

moves expensive candidate batches, NCA training, ART2/ARTMAP discovery, perturbation sweeps, and result return onto remote compute

The scientific guardrail stays the same:

A remote run is not successful because it made a nice image.

A remote run is successful if it returns inspectable mechanism evidence:

dynamics metrics

NCA agreement

ART2 prototypes

ARTMAP transitions

perturbation response

DSL artifacts

mechanism reports

This stays aligned with Hiscock/Megason: final pattern similarity alone is weak evidence; runs should preserve dynamics, perturbation responses, parameter constraints, and mechanism-discrimination information.

Where this bundle sits in the platform

Bundle 5 is still mostly **config-managed runtime preparation**, plus tiny execution smoke tests.

config

prepares Runpod workspace paths, job layouts, checkpoint policy, result-return policy, and smoke commands

nca-art-grn repo

owns training/evaluation/search CLI code

/workspace

stores inputs, runs, artifacts, checkpoints, models

Runpod

executes expensive jobs

Agentfield later

schedules Runpod-backed experiments/campaigns

Paperclip later

shows job status, costs, artifacts, and review reports

These are not Agentfield controller steps yet. They are the **Runpod execution foundation** that Agentfield will later call.

User story

As the Research Scientist / Operator,
I want a prepared Runpod execution loop for NCA-ART-GRN experiments,
so that candidate batches, NCA training, ART2/ARTMAP discovery, perturbation sweeps,
and mechanism reports can run remotely, return results safely, and remain comparable
with local smoke runs.

Concretizations / managed steps

```
prepare_runpod_training_workspace  
prepare_runpod_inference_workspace  
prepare_candidate_batch_layout  
prepare_training_run_layout  
prepare_checkpoint_policy  
prepare_result_return_policy  
prepare_remote_run_manifest_schema  
prepare_runpod_job_templates  
prepare_runpod_nca_training_configs  
prepare_runpod_art_discovery_configs  
prepare_runpod_search_campaign_configs  
prepare_runpod_mechanism_report_configs  
check_runpod_training_ready  
run_runpod_local_dryrun_smoke
```

Optional later:

```
submit_runpod_training_job  
pull_runpod_results  
resume_runpod_checkpoint
```

Those should be explicit later commands, not automatic prepare steps.

Step type legend

CONFIG STEP

Managed bootstrap/check/smoke step invoked by config.

REPO CODE

Python modules, CLIs, configs, tests, scripts inside nca-art-grn.

REMOTE EXECUTION

Runpod job or container execution.

RESEARCH OUTPUT

Candidate batches, run outputs, checkpoints, artifacts, reports.

AGENTFIELD LATER

Future orchestration layer; not implemented in this bundle.

What each step should do

`prepare_runpod_training_workspace`

Type: CONFIG STEP

Owner: config prepares remote-training paths.

Target role: usually researchscientist .

Creates shared paths:

```
/workspace/runs/nca-art-grn/runpod/  
/workspace/checkpoints/nca-art-grn/  
/workspace/models/nca-art-grn/
```

/workspace/artifacts/nca-art-grn/runpod/

What this does for the research:

Gives expensive NCA training and large simulation/search runs a stable place to write outputs without polluting the repo.

How it links into the platform:

```
config prepares paths
repo training CLI writes there
Runpod mounts /workspace
Agentfield later stores these paths in experiment status
Paperclip later displays returned run artifacts
```

prepare_runpod_inference_workspace

Type: CONFIG STEP

Creates paths for remote evaluation/inference:

```
/workspace/runs/nca-art-grn/inference/
/workspace/artifacts/nca-art-grn/inference/
/workspace/models/nca-art-grn/served/
/workspace/data/nca-art-grn/inference_inputs/
```

What this does for the research:

Separates long training from later evaluation/inference. For example, after an NCA model is trained, you can run rollout evaluation, perturbation replay, ART2 prototype extraction, or candidate scoring as separate jobs.

How it links into the platform:

```
trained model checkpoint -> inference workspace
inference run -> dynamics/prototype/transition artifacts
```

Agentfield later can schedule inference only experiments

prepare_candidate_batch_layout

Type: CONFIG STEP

Creates research input layout:

```
/workspace/data/nca-art-grn/candidate_batches/  
  pending/  
  active/  
  completed/  
  failed/
```

Each batch should contain:

```
batch_manifest.yaml  
candidates/  
  candidate_000001.dsl.json  
  candidate_000002.dsl.json  
configs/  
  evaluation_config.yaml  
  perturbation_config.yaml  
metadata.json
```

The batch manifest must encode:

```
batch_id  
created_by  
candidate_count  
source_search_method  
mechanism_hypothesis_ids  
simulation_budget  
nca_budget
```

```
art2_settings
artmap_settings
perturbation_plan
expected outputs
```

What this does for the research:

Lets you send a controlled set of DSL candidates to remote compute. This keeps remote jobs traceable to search method, mechanism hypothesis, and perturbation plan.

How it links into the platform:

```
Bundle 4 produces ranked/search candidates
candidate_batch_layout packages them
Runpod evaluates batch
results return to /workspace/runs and /workspace/artifacts
Agentfield later treats batch_manifest as experiment input
```

prepare_training_run_layout

Type: CONFIG STEP

Creates standard run directory schema:

```
/workspace/runs/nca-art-grn/runpod/<run_id>/
run_manifest.yaml
environment.json
command.txt
inputs/
logs/
checkpoints/
outputs/
artifacts/
reports/
status.json
```

run_manifest.yaml must encode:

```
run_id
batch_id
repo_commit_or_snapshot
target_role
python_env
container_image
gpu_type
started_at
command
input_paths
output_paths
checkpoint_paths
cost_policy
failure_policy
```

What this does for the research:

Makes remote runs reproducible and auditable. A run is not just “something happened on Runpod”; it becomes a structured scientific execution record.

How it links into the platform:

```
config creates run layout
repo writes outputs into run layout
Agentfield later maps run_manifest to experiment status
Paperclip later displays run lifecycle and failure reason
```

prepare_checkpoint_policy

Type: CONFIG STEP**Creates policy/config files:**

```
configs/runpod/checkpoint_policy.yaml  
/workspace/checkpoints/nca-art-grn/README.md
```

Policy must encode:

```
checkpoint frequency  
checkpoint naming  
keep-last-N policy  
best-model policy  
resume policy  
failed-run recovery policy  
large-file handling  
model artifact promotion rules
```

What this does for the research:

Protects expensive NCA training and long evaluations from being lost. It also prevents uncontrolled checkpoint sprawl.

How it links into the platform:

```
NCA training writes checkpoints  
inference jobs consume promoted checkpoints  
Agentfield later resumes failed jobs  
Paperclip later exposes checkpoint status
```

```
prepare_result_return_policy
```

Type: CONFIG STEP**Creates policy/config files:**

```
configs/runpod/result_return_policy.yaml  
/workspace/artifacts/nca-art-grn/result_return/README.md
```

Policy must specify which outputs are returned/promoted:

```
metadata.json  
run_manifest.yaml  
candidate DSL files  
pattern_dynamics.json  
nca_summary.json  
art2_prototypes.json  
artmap_transitions.json  
perturbation_summary.json  
mechanism_report.md  
search_report.md  
figures/  
tables/  
logs summary  
failure_reason.txt
```

It should avoid returning uncontrolled huge files by default.

What this does for the research:

Ensures that remote results are useful for science, not just raw dumped files. The key returned artifact is the mechanism evidence trail.

How it links into the platform:

```
Runpod writes outputs  
result_return_policy selects important artifacts  
Bundle 9 can turn reports into notes  
Bundle 10 can use figures/tables/reports  
Paperclip later shows selected outputs
```

prepare_remote_run_manifest_schema

Type: CONFIG STEP

Creates/validates repo schema:

```
src/nca_art_grn/runs/remote_manifest.py  
configs/runpod/remote_run_manifest_schema.yaml
```

Schema must include:

```
run_id  
run_kind  
candidate_batch_id  
mechanism_hypothesis_ids  
search_method  
nca_training_mode  
art2_mode  
artmap_mode  
perturbation_plan_id  
input_paths  
output_paths  
checkpoint_policy  
result_return_policy  
cost_limit  
expected_artifacts
```

What this does for the research:

Makes local, Runpod, and later Agentfield runs speak the same language.

How it links into the platform:

```
config writes manifest  
repo CLI reads manifest
```



Runpod executes manifest
Agentfield later creates manifest from GRNExperiment spec
Paperclip later shows manifest summary

prepare_runpod_job_templates

Type: CONFIG STEP

Creates scripts/templates:

```
scripts/runpod/run_batch.sh
scripts/runpod/run_nca_training.sh
scripts/runpod/run_art_discovery.sh
scripts/runpod/run_search_campaign.sh
scripts/runpod/collect_results.sh
configs/runpod/job_templates/
```

Each template must define:

```
working directory
environment activation
input manifest path
output directory
logging path
failure trap
exit code handling
result collection
```

What this does for the research:

Provides repeatable remote commands without having to remember ad hoc shell incantations.

How it links into the platform:

operator can run template manually
config can dry-run template
Agentfield later calls same template from controller
Paperclip later displays template/job kind

prepare_runpod_nca_training_configs

Type: CONFIG STEP

Creates configs:

configs/runpod/nca_training_small.yaml
configs/runpod/nca_training_medium.yaml
configs/runpod/nca_training_resume.yaml

Config must encode:

dataset path
candidate batch path
model config
loss config
training steps
checkpoint interval
validation interval
rollout length
perturbation validation
device policy

What this does for the research:

Lets NCA training scale from tiny local smoke to real remote training while preserving the same data contract from Bundle 3.

How it links into the platform:

PDE/ODE-to-NCA datasets -> NCA training config
training -> checkpoints
checkpoints -> inference workspace
Agentfield later schedules NCA training experiments

prepare_runpod_art_discovery_configs

Type: CONFIG STEP

Creates configs:

configs/runpod/art2_discovery_batch.yaml
configs/runpod/artmap_transition_batch.yaml
configs/runpod/prototype_to_dsl_batch.yaml

Config must encode:

input trajectory paths
NCA rollout paths
ART2 vigilance settings
ARTMAP mapping settings
prototype store target
transition graph target
mechanism report target

What this does for the research:

Lets ART2 and ARTMAP run on larger trajectory collections than local smoke can handle.

How it links into the platform:

remote simulator/NCA outputs -> ART2/ARTMAP batch
ART2/ARTMAP -> prototypes and transitions

prototype_to_dsl -> DSL artifacts

prepare_runpod_search_campaign_configs

Type: CONFIG STEP

Creates configs:

```
configs/runpod/search_random_grid_campaign.yaml
configs/runpod/search_lhs_campaign.yaml
configs/runpod/search_evolutionary_campaign.yaml
configs/runpod/search_bayesian_campaign.yaml
configs/runpod/search_active_learning_campaign.yaml
```

Config must encode:

```
search method
candidate budget
remote batch size
simulation budget per candidate
NCA evaluation budget
ART2/ARTMAP evaluation budget
perturbation budget
checkpoint policy
result return policy
cost guardrails
```

What this does for the research:

Moves Bundle 4's search comparison from toy local smoke into actual remote campaigns.

How it links into the platform:

```
Bundle 4 search config -> Runpod campaign config
Runpod campaign -> ranked candidates and reports
```

Agentfield later tracks campaign phase/status
Paperclip later enables human review of campaign outputs

prepare_runpod_mechanism_report_configs

Type: CONFIG STEP

Creates configs/templates:

configs/runpod/mechanism_report_batch.yaml
configs/reports/runpod_mechanism_report_template.md

Report must summarize:

candidate batch
best candidates
failed candidates
dynamics evidence
NCA agreement
ART2 prototype evidence
ARTMAP transition evidence
perturbation evidence
DSL mapping evidence
experimental design suggestions
cost/runtime summary

What this does for the research:

Ensures remote campaigns return science-ready interpretation, not just raw logs.

How it links into the platform:

remote outputs -> mechanism reports
reports -> Bundle 9 alloy notes

```
reports -> Bundle 10 paper sections  
reports -> Agentfield status artifact  
reports -> Panacrin review item
```

check_runpod_training_ready

Type: CONFIG CHECK STEP

Runs: no training, no Runpod launch.

Should report:

```
workspace paths exist  
repo exists  
Python env exists  
GPU visible if on Runpod  
CUDA/Torch ready  
candidate batch path exists  
checkpoint path exists  
result return path exists  
runpod configs exist  
job templates exist  
required package imports available  
remote model/client env optional status
```

What this does for the research:

Prevents expensive failed remote jobs caused by missing paths, packages, configs, or GPU visibility.

How it links into the platform:

```
operator runs check  
researchscientist fixes missing pieces  
Agentfield later runs preflight before scheduling remote job
```

run_runpod_local_dryrun_smoke

Type: CONFIG SMOKE STEP

Runs: local dry-run only; no paid remote launch.

Should do:

```
create fake run_id
load tiny remote manifest
validate candidate batch manifest
validate job template command
validate output directories
write dryrun status.json
write dryrun report
```

Output:

```
/workspace/runs/nca-art-grn/runpod/dryrun/<timestamp>/
run_manifest.yaml
command.txt
status.json
dryrun_report.md
```

What this does for the research:

Confirms the Runpod execution contract without spending money or launching a job.

How it links into the platform:

```
today:
  config validates remote-run shape

later:
  submit_runpod_training_job uses same manifest
```

later:

Whole-system linkage

Bundle 5 is the bridge from local research engine to remote execution.

Bundle 3

defines and evaluates mechanism candidates

Bundle 4

chooses candidate/search campaigns

Bundle 5

executes those campaigns remotely and returns structured evidence

Bundle 11 later

Agentfield controls experiment lifecycle

Bundle 12 later

Paperclip-Agentfield adapter exposes that lifecycle to Paperclip

Paperclip later

shows remote job status, artifacts, reports, and review actions

Concrete later flow:

config today:

```
sudo config --target researchscientist bootstrap step run_runpod_local_dryrun_smoke
```

manual repo command:

```
python -m nca_art_grn.cli.remote_dryrun \  
--manifest /workspace/runs/nca-art-grn/runpod/dryrun/<id>/run_manifest.yaml
```

future Agentfield:

```
GRNExperiment.spec.execution.target = runpod
GRNExperiment.spec.method = nca_art_pipeline | parameter_search | nca_training
controller writes run_manifest.yaml
controller launches Runpod job
status points to /workspace/runs/nca-art-grn/runpod/<run_id>
```

future Paperclip:

```
displays queued/running/failed/completed
links mechanism_report.md, search_report.md, prototypes, figures, checkpoints
```

Acceptance criteria

The following should become valid:

```
sudo config --target researchscientist bootstrap step prepare_runpod_training_workspace
sudo config --target researchscientist bootstrap step prepare_runpod_inference_workspace
sudo config --target researchscientist bootstrap step prepare_candidate_batch_layout
sudo config --target researchscientist bootstrap step prepare_training_run_layout
sudo config --target researchscientist bootstrap step prepare_checkpoint_policy
sudo config --target researchscientist bootstrap step prepare_result_return_policy
sudo config --target researchscientist bootstrap step prepare_remote_run_manifest_schema
sudo config --target researchscientist bootstrap step prepare_runpod_job_templates
sudo config --target researchscientist bootstrap step prepare_runpod_nca_training_configs
sudo config --target researchscientist bootstrap step prepare_runpod_art_discovery_configs
sudo config --target researchscientist bootstrap step prepare_runpod_search_campaign_configs
sudo config --target researchscientist bootstrap step prepare_runpod_mechanism_report_configs
sudo config --target researchscientist bootstrap step check_runpod_training_ready
```

Explicit dry-run:

```
sudo config --target researchscientist bootstrap step run_runpod_local_dryrun_smoke
```

Prepare/check steps must:

```
not launch Runpod jobs
not train models
not run real campaigns
not overwrite checkpoints
not overwrite completed runs
not print secrets
not upload private data automatically
not delete remote results
```

Dry-run step must:

```
validate manifests
validate paths
validate command templates
write dryrun report
clearly label outputs as dryrun
not use GPU
not call paid remote APIs
not submit jobs
```

Proposed tests

Registry and syntax tests

```
bash --noprofile --norc -n /home/vmuser/.local/bin/config.sh
bash --noprofile --norc -n /home/vmuser/.local/lib/config-sh/installers.sh
```



```
config bootstrap steps | grep prepare_runpod_training_workspace
config bootstrap steps | grep prepare_candidate_batch_layout
config bootstrap steps | grep prepare_checkpoint_policy
config bootstrap steps | grep prepare_result_return_policy
config bootstrap steps | grep prepare_runpod_job_templates
config bootstrap steps | grep check runpod training ready
```

Workspace structure tests

```
sudo config --target researchscientist bootstrap step prepare_runpod_training_workspace
sudo config --target researchscientist bootstrap step prepare_runpod_inference_workspace
```

```
test -d /workspace/runs/nca-art-grn/runpod
test -d /workspace/checkpoints/nca-art-grn
test -d /workspace/models/nca-art-grn
test -d /workspace/artifacts/nca-art-grn/runpod
```

Candidate batch tests

```
sudo config --target researchscientist bootstrap step prepare_candidate_batch_layout
```

```
test -d /workspace/data/nca-art-grn/candidate_batches/pending
test -d /workspace/data/nca-art-grn/candidate_batches/active
test -d /workspace/data/nca-art-grn/candidate_batches/completed
test -d /workspace/data/nca-art-grn/candidate_batches/failed
```

Policy tests

```
sudo config --target researchscientist bootstrap step prepare_checkpoint_policy
sudo config --target researchscientist bootstrap step prepare_result_return_policy
```

```
test -f /workspace/repos/nca-art-grn/configs/runpod/checkpoint_policy.yaml
test -f /workspace/repos/nca-art-grn/configs/runpod/result_return_policy.yaml
```

Job template tests

```
sudo config --target researchscientist bootstrap step prepare_runpod_job_templates

test -f /workspace/repos/nca-art-grn/scripts/runpod/run_batch.sh
test -f /workspace/repos/nca-art-grn/scripts/runpod/run_nca_training.sh
test -f /workspace/repos/nca-art-grn/scripts/runpod/run_art_discovery.sh
test -f /workspace/repos/nca-art-grn/scripts/runpod/run_search_campaign.sh

bash --noprofile --norc -n /workspace/repos/nca-art-grn/scripts/runpod/run_batch.sh
bash --noprofile --norc -n /workspace/repos/nca-art-grn/scripts/runpod/run_nca_training.sh
```

Readiness check

```
sudo config --target researchscientist bootstrap step check_runpod_training_ready
```

Expected: readiness report only, no job launch.

Local dry-run smoke

```
sudo config --target researchscientist bootstrap step run_runpod_local_dryrun_smoke

find /workspace/runs/nca-art-grn/runpod/dryrun -name run_manifest.yaml | tail -n 1
find /workspace/runs/nca-art-grn/runpod/dryrun -name status.json | tail -n 1
find /workspace/runs/nca-art-grn/runpod/dryrun -name dryrun_report.md | tail -n 1
```

Scientific output-contract test

The result-return policy must include:

```
candidate DSL files
pattern_dynamics.json
nca_summary.json
art2_prototypes.json
artmap_transitions.json
perturbation_summary.json
mechanism_report.md
search_report.md
```

Non-overwrite checkpoint test

```
mkdir -p /workspace/checkpoints/nca-art-grn/test_checkpoint
echo "DO NOT OVERWRITE" > /workspace/checkpoints/nca-art-grn/test_checkpoint/marker.txt
sudo config --target researchscientist bootstrap step prepare_checkpoint_policy
grep "DO NOT OVERWRITE" /workspace/checkpoints/nca-art-grn/test_checkpoint/marker.txt
```

Summary

Bundle 5 is the **remote execution and result-return layer**.

It does not replace Bundle 3 or 4.

Bundle 3:

makes one mechanism-evaluation loop executable

Bundle 4:

makes many candidates/search methods comparable

Bundle 5:

makes expensive versions run remotely and return structured evidence

Scientifically, it must return:

not just images,
but candidate DSL,
dynamics metrics,
NCA summaries,
ART2 prototypes,
ARTMAP transitions,
perturbation results,
mechanism reports,
search reports,
figures,
tables,
and failure reasons.

Platform-wise:

config prepares remote execution contracts
nca-art-grn owns the executable research code
/workspace stores all remote inputs/outputs
Agentfield later schedules these jobs
Paperclip later lets you review them